

Compensated Positive Kernels for Polylogarithms

Subcritical zero geometry, a sharp critical Euler constant,
and exact generalized Stieltjes order at every depth

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Abstract

For $a, b > 0$, put $F_{a,b}(z) = \sum_{n \geq 1} H_{n-1}^{(b)} z^n / n^a$. The preceding real-order report proves that its coefficients have a finite signed Hausdorff representation precisely when $a + b \geq 1$, and leaves angular uniqueness below this boundary and a sharp critical Euler constant as explicit conjectures. We prove both conjectures.

When $a + b \leq 1$, an explicit positive, generally infinite measure ν represents the coefficients through the compensated moments $c_n = \int (1 - u^{n-1}) d\nu(u)$. Its finite first-difference measure gives $F_{a,b}(z) = z^2(1 - z)^{-1} \int (1 - zu)^{-1} (1 - u) d\nu(u)$. This proves slit-plane nonvanishing below the old threshold, unique simple angular zeros, and strict increase of the normalized radial displacement throughout the closed subcritical region. Combined with the prior supercritical theorem, nonvanishing and angular uniqueness hold for all positive orders. A beta likelihood-ratio argument proves that $-\operatorname{Im} F_{a,w-a}(i\rho)$ decreases strictly with a for every fixed $w > 0$ and $0 < \rho \leq 1$. On $w = 1$ this yields the conjectured optimal uniform Euler constant $\pi/4 + (\log 2)/2$.

A separate positive transport formula supplies signed Euler differences and explicit geometric remainders at every positive pair, including where no finite signed measure exists. At arbitrary strict nesting depth d , we construct a positive generalized Stieltjes representation of order d and prove that this order is exact, using a logarithmic negative-axis asymptotic. Equivalently, the coefficient normalization $c_{n+d} n! / (\alpha)_n$ is a positive Hausdorff moment sequence exactly for $\alpha \geq d$.

The package includes complete ordinary mathematical proofs, exact rational certificates for nine Gaussian values and twelve subcritical angular brackets, symbolic checks, and integration notes. The results are not proof-assistant formalizations or independently refereed claims. The S_6 period identity and the general supercritical normalized-radius problem remain unresolved here.

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1 Scope, sources, and principal contributions

1.1 The concrete questions being continued

This article continues the canonical manuscript [1] at revision `a0a90ef31877f98be437191c48b46f02d5456867`, together with the incoming report *The Sharp Real-Order Threshold for Harmonic Polylogarithms* [2]. The latter is particularly important: its conjectures are later than several already consolidated results. It explicitly asks whether the angular zero remains unique when $a + b < 1$, whether the slit-plane nonvanishing theorem extends to that region, and whether $a \mapsto -2 \operatorname{Im} F_{a,1-a}(i)$ decreases strictly with sharp supremum $\pi/4 + (\log 2)/2$. These are the questions answered below.

The previous report already proves the finite signed-measure threshold, the critical endpoint atom, critical normalized-radius monotonicity, and the opposite behavior of $\operatorname{Li}_a(z) - z$. Its positive two-variable kernel is also prior material. We use these insights, rederive the needed formulas, and do not present them as discoveries of this article. The canonical manuscript already has exact S_4 proofs and retains S_6 as a conjectural mixed period reduction. Neither status is changed here.

The inspection was targeted, not a line-by-line audit of every source package. The incoming directory was inventoried through GitHub; the threshold report was read from its accessible Library source. Its source hash and the distinction between that copy and the repository archive are recorded in `PROVENANCE.json`. The supplied package does not modify a repository branch.

Question	Inspected prior result	Result proved here
Finite signed moments	Exactly $a + b \geq 1$	Threshold retained; compensated positive moments fill $a + b < 1$
Angular zero	Unique for $a + b \geq 1$	Unique and simple for all $a, b > 0$
Principal-branch zeros	Only the double zero at zero for $a + b \geq 1$	Same statement for all $a, b > 0$
Normalized radius	Increasing on $a + b = 1$	Increasing on all $a + b \leq 1$
Critical Euler constant	Explicit conjecture	Sharp constant $\pi/4 + (\log 2)/2$
All-order Euler evaluation	Finite-measure proof on $a + b \geq 1$	Positive transport proof at all positive orders
Strict depth d	No such result in the inspected continuation	Exact Stieltjes order d ; sharp normalization threshold

1.2 Conventions and the meaning of exactness

Throughout,

$$H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad c_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad F_{a,b}(z) = \sum_{n \geq 1} c_n(a, b) z^n. \quad (1.1)$$

We take $H_0^{(b)} = 0$, so $c_1 = 0$ and $c_2 = 2^{-a}$. The initial series is used on $|z| < 1$ and continued to

$$\Omega = \mathbb{C} \setminus [1, \infty).$$

All indices in this article are real and positive unless a boundary value is expressly specified. Nested sums use strictly decreasing indices, not the star convention. Depth-one branch conventions agree with DLMF §25.12 [4]. We abbreviate

$$w = a + b, \quad g_{a,b}(\rho) = \operatorname{Im} F_{a,b}(i\rho), \quad g_{a,b} = g_{a,b}(1).$$

Root-of-unity values on $w \leq 1$ mean analytic, equivalently Abel, values. The original series then fails the ordinary term test.

A proved integral identity, a finite exact-arithmetic test, and a floating-point diagnostic are different forms of evidence. Every theorem below has a written proof. The accompanying tests check finite algebraic identities, value enclosures, and implementation behavior; they do not replace the proofs. No numerical agreement is used to prove a period identity, and no statement of numerical period independence is made. The novelty assessment is relative to the inspected project sources, not a claim of exhaustive priority over the literature.

2 Positive kernels and two different moment thresholds

Set

$$k_s(T) = \frac{T^{s-1}}{\Gamma(s)}, \quad h(T) = \frac{1}{1 - e^{-T}} - \frac{1}{T} \quad (T > 0). \quad (2.1)$$

The gamma integral gives $\int_0^\infty e^{-qT} k_s(T) dT = q^{-s}$ for $q, s > 0$. The function h increases strictly from $1/2$ to 1 , because

$$h'(T) = \frac{1}{T^2} - \frac{1}{4 \sinh^2(T/2)} > 0.$$

Define

$$C_{a,b}(T) = \int_0^T k_a(T-t) k_b(t) h(t) dt \quad (2.2)$$

$$= k_w(T) \mathbb{E}[h(TV)], \quad V \sim \text{Beta}(b, a). \quad (2.3)$$

In particular $\frac{1}{2}k_w < C_{a,b} < k_w$.

Lemma 2.1 (Regularized Hurwitz transform). *For $q > 0$, $b > 0$, $b \neq 1$,*

$$\zeta(b, q) = \frac{q^{1-b}}{b-1} + \int_0^\infty e^{-qt} k_b(t) h(t) dt. \quad (2.4)$$

Consequently

$$\int_0^\infty e^{-qT} C_{a,b}(T) dT = q^{-a} \left(\zeta(b, q) - \frac{q^{1-b}}{b-1} \right). \quad (2.5)$$

At $b = 1$, the parenthesis is interpreted as $\log q - \psi(q)$.

Proof. For $b > 1$, split the classical Hurwitz integral into $1/t + h(t)$. The first part is $q^{1-b}/(b-1)$. Since h is bounded, the other integral is locally holomorphic for $\text{Re } b > 0$. Analytic continuation in b proves the asserted formula away from its pole. This is the regularization recorded in DLMF §25.11 [3]. Taking constant terms at $b = 1$ gives $\log q - \psi(q)$. Tonelli's theorem and the gamma integral prove the convolution formula. $\square$

Lemma 2.2 (A positive double kernel; prior construction). *For every $a, b > 0$ and $z \in \Omega$,*

$$F_{a,b}(z) = z^2 \int_{0 < y < x < 1} \frac{W_{a,b}(x, y)}{(1 - zx)(1 - zy)} dy dx, \quad (2.6)$$

where

$$W_{a,b}(x, y) = \frac{(-\log x)^{a-1} (\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)}. \quad (2.7)$$

This weight is positive and has total mass 2^{-a} .

Proof. For nonnegative integers j, k , substitution $y = xv$ gives

$$\int W_{a,b}(x, y) x^j y^k dy dx = \frac{1}{(j+k+2)^a (k+1)^b}.$$

Expanding the two denominators in the disk and summing over $j+k = n-2$ therefore gives c_n . Absolute convergence justifies the expansion. On a compact subset of Ω , $1-zx$ and $1-zy$ stay uniformly away from zero for $0 \leq x, y \leq 1$. The finite integral is holomorphic there and supplies the continuation. Take $j = k = 0$ for its mass. $\square$

The following preceding result is included to make the all-order conclusions independent of an unquoted sign argument.

Proposition 2.3 (The finite signed threshold; prior theorem). *There is a finite real signed measure μ on $[0, 1]$ satisfying $c_n = \int u^{n-1} d\mu(u)$ for all $n \geq 1$ if and only if $w \geq 1$. For $w > 1$ it has density $K_{a,b}(-\log u)$, where*

$$K_{a,b}(T) = \zeta(b)k_a(T) - \frac{k_{w-1}(T)}{b-1} - C_{a,b}(T) \quad (b \neq 1), \quad (2.8)$$

$$K_{a,1}(T) = k_a(T)(\gamma + \psi(a) - \log T) - C_{a,1}(T). \quad (2.9)$$

This density has exactly one sign change, negative to positive as u increases. On $w = 1$, the measure instead has the form $\mu = a^{-1}\delta_1 - \nu$ with ν a positive finite measure.

Proof. The densities in (2.8)–(2.9) are integrable with weight $e^{-T} dT$. Their Laplace transforms are c_n by Lemma 2.1; at $b = 1$, also use

$$\int_0^\infty e^{-qT} k_a(T) \log T dT = q^{-a}(\psi(a) - \log q).$$

In particular the signed measure has total mass zero. For $b \neq 1$, differentiation after dividing by $k_a(T)$ gives

$$\begin{aligned} \left(\frac{K_{a,b}}{k_a} \right)'(T) &= -\frac{\Gamma(a)}{\Gamma(w-1)} T^{b-2} \\ &\quad - \frac{\Gamma(a)}{\Gamma(w)} T^{b-1} \mathbb{E}[bh(TV) + TVh'(TV)] < 0. \end{aligned}$$

For $b = 1$ the derivative is $-T^{-1} - a^{-1}\mathbb{E}[h(TV) + TVh'(TV)] < 0$. The normalized density starts positive (possibly at $+\infty$) and tends to $-\infty$. This proves exactly one transversal crossing, with the stated orientation after $u = e^{-T}$.

The construction on $w = 1$ follows from Theorem 3.1 below, where ν has mass $1/a$. For $w < 1$, the integral test gives $c_n \sim n^{1-w}/(1-b) \rightarrow \infty$, impossible for moments of a finite signed measure on $[0, 1]$. Uniqueness in the finite cases follows from density of polynomials in the continuous functions on $[0, 1]$. $\square$

3 The compensated subcritical measure

The absence of a finite signed measure does not preclude a positive representation of *differences*. Here that distinction is decisive.

Theorem 3.1 (Explicit positive compensation). *Let $a, b > 0$ and $w = a + b \leq 1$. Put*

$$\delta = 1 - w \in [0, 1), \quad A = \frac{1}{1-b}, \quad N_{a,b}(T) = -\zeta(b)k_a(T) + C_{a,b}(T).$$

Define a positive measure on $(0, 1)$ by

$$d\nu_{a,b}(u) = \left[N_{a,b}(-\log u) + \frac{\delta}{(1-b)\Gamma(1-\delta)} (-\log u)^{-1-\delta} \right] du. \quad (3.1)$$

When $\delta = 0$, the second summand is omitted. Then

$$c_n = \int_0^1 (1 - u^{n-1}) d\nu_{a,b}(u), \quad n \geq 1, \quad (3.2)$$

and

$$d\omega_{a,b}(u) = (1 - u) d\nu_{a,b}(u), \quad \omega_{a,b}((0, 1)) = 2^{-a}. \quad (3.3)$$

The density of $\omega_{a,b}$ is strictly positive throughout $(0, 1)$. The measure $\nu_{a,b}$ is finite precisely on $w = 1$, where its mass is $A = 1/a$. For $w < 1$, its total mass is infinite, but every compensated integral in (3.2) is finite.

Proof. Since $0 < b < 1$, $\zeta(b) < 0$. For example, the alternating Dirichlet series is positive and $\zeta(b) = \eta_D(b)/(1 - 2^{1-b})$. Thus $N_{a,b} > 0$. Lemma 2.1 gives, for $q > 0$,

$$q^{-a}(\zeta(b) - \zeta(b, q)) = Aq^\delta - \int_0^\infty e^{-qT} N_{a,b}(T) dT. \quad (3.4)$$

At $q = 1$ the left side is zero, so $\int e^{-T} N_{a,b}(T) dT = A$. For $0 < \delta < 1$, integration by parts gives

$$q^\delta - 1 = \frac{\delta}{\Gamma(1-\delta)} \int_0^\infty (e^{-T} - e^{-qT}) T^{-1-\delta} dT. \quad (3.5)$$

Indeed the boundary term vanishes at both endpoints, and the remaining gamma integrals evaluate the expression. Subtract the $q = 1$ identity from (3.4), insert (3.5), and change variables $u = e^{-T}$. For integer $q = n$, the Hurwitz difference is $H_{n-1}^{(b)}$, proving (3.2). At $\delta = 0$ the same argument works without the fractional-power term.

Near $u = 1$, the new density has leading behavior

$$\frac{\delta}{(1-b)\Gamma(1-\delta)} (1-u)^{-1-\delta} \quad (\delta > 0).$$

Its uncompensated mass diverges, while multiplication by $1 - u$ makes it integrable because $\delta < 1$. The remaining density is integrable: its e^{-T} -weighted mass is A . At $u = 0$ exponential decay in the T coordinate makes all terms integrable. Setting $n = 2$ gives the mass in (3.3). Strict positivity is immediate. $\square$

Remark 3.2 (The cancellation is mathematical, not notational). For $w < 1$, it is invalid to rewrite $\int (1 - u^{n-1}) d\nu$ as the difference of its two separate integrals: both are infinite. Nor is $\infty\delta_1 - \nu$ a finite signed measure. The finite object to use for zero geometry is $\omega = (1 - u)\nu$. This retains, rather than contradicts, the threshold of Proposition 2.3.

Corollary 3.3 (A positive first-difference Stieltjes factor). *For $w \leq 1$ and $z \in \Omega$,*

$$F_{a,b}(z) = \frac{z^2}{1-z} P_{a,b}(z), \quad P_{a,b}(z) = \int_0^1 \frac{d\omega_{a,b}(u)}{1-zu}. \quad (3.6)$$

Equivalently,

$$\frac{(1-z)F_{a,b}(z)}{z^2} = \sum_{n \geq 0} (c_{n+2} - c_{n+1})z^n. \quad (3.7)$$

The removable value at zero is 2^{-a} .

Proof. In the disk,

$$\sum_{n \geq 1} (1 - u^{n-1})z^n = \frac{z^2(1-u)}{(1-z)(1-zu)}.$$

Termwise integration is justified by $|1 - u^{n-1}| \leq (n-1)(1-u)$ and finiteness of ω . Both sides continue holomorphically to Ω , proving the claim. $\square$

Theorem 3.4 (A sharp complementary positivity threshold). *For $a, b > 0$, the sequence $(c_{n+2} - c_{n+1})_{n \geq 0}$ is a positive Hausdorff moment sequence if and only if $a + b \leq 1$. In that region, with $\nabla f_n = f_{n+1} - f_n$,*

$$(-1)^{k-1} \nabla^k c_{n+1} = \int_0^1 u^n (1-u)^k d\nu_{a,b}(u) > 0 \quad (n \geq 0, k \geq 1). \quad (3.8)$$

All finite principal Hankel matrices of the first-difference sequence are positive definite. The continuous function

$$B_{a,b}(x) = (x+1)^{-a} (\zeta(b) - \zeta(b, x+1)), \quad x \geq 0, \quad (3.9)$$

is a Bernstein function: it is nonnegative and its derivative is strictly completely monotone.

Proof. Equation (3.8) follows by taking finite differences inside (3.2). Its integrals are finite because $k \geq 1$. The case $k = 1$ identifies the first-difference sequence as the moments of ω . For a nonzero polynomial p , $\int p(u)^2 d\omega(u) > 0$, since ω has positive density on every interior interval. This proves strict Hankel positivity.

When $w > 1$, $c_n \rightarrow 0$: for $b < 1$ use the integral test, for $b = 1$ use $H_{n-1} = O(\log n)$, and for $b > 1$ use boundedness of the harmonic sums. Since $c_1 = 0 < c_2$, the differences cannot all be nonnegative. They therefore cannot be moments of a positive measure. This proves necessity, not just a sufficient construction.

The proof of Theorem 3.1 also works for $q = x+1$, giving $B(x) = \int (1-u^x) d\nu(u)$. Differentiation for $x > 0$ yields

$$(-1)^j B^{(j+1)}(x) = \int_0^1 u^x (-\log u)^{j+1} d\nu(u) > 0.$$

Near one the integrability condition is again $\delta < 1$, and near zero the u density supplies exponential decay. The same domination also gives continuity at $x = 0$. $\square$

4 All-positive-order nonvanishing and angular uniqueness

Theorem 4.1 (No nonzero zeros on the principal slit plane). *For every $a, b > 0$, the only zero of $F_{a,b}$ on Ω is its double zero at 0.*

Proof. For $w \leq 1$, use (3.6). If $\operatorname{Im} z \neq 0$, then

$$\operatorname{Im} P_{a,b}(z) = \operatorname{Im} z \int_0^1 \frac{u \, d\omega_{a,b}(u)}{|1 - zu|^2},$$

whose integral is strictly positive. For real $z < 1$, $P_{a,b}(z) > 0$. The factor $1 - z$ has no zero on Ω , and $P_{a,b}(0) = 2^{-a}$.

For completeness, when $w > 1$, let $r \in (0, 1)$ be the crossing of Proposition 2.3. The positive nonzero measure $\lambda = (u - r)\mu$ gives

$$F_{a,b}(z) = \frac{z^2}{1 - rz} \int_0^1 \frac{d\lambda(u)}{1 - zu}.$$

This follows by subtracting $z(1 - rz)^{-1} \int d\mu = 0$ from its finite signed-measure representation. The same imaginary-part test proves nonvanishing of the integral, and $1 - rz$ has no zero on Ω . The coefficient $c_2 = 2^{-a}$ gives exact multiplicity two. $\square$

4.1 A root lemma that permits infinite uncompensated mass

Lemma 4.2 (Endpoint factor and normalized root). *Let ω be a finite positive measure on $(0, 1)$ with positive density there. Define*

$$F(z) = \frac{z^2}{1 - z} \int_0^1 \frac{d\omega(u)}{1 - zu}.$$

For every $0 < \rho \leq 1$, $\operatorname{Im} F(\rho e^{i\theta})$ has exactly one zero in $0 < \theta < \pi$. It is simple, has positive sign before and negative sign after the zero, and satisfies

$$\arccos \rho < \theta(\rho) < \arccos(\rho/2). \quad (4.1)$$

Moreover $\eta(\rho) = \cos \theta(\rho)/\rho$ increases strictly.

Proof. Write

$$x = \rho^2, \quad c = \cos \theta, \quad m = \frac{2c}{\rho} - 1, \quad D(u) = 1 + x(u^2 - (1 + m)u).$$

For the relevant root range $0 < m < 1$, all denominators are positive, including $D(1) = 1 - xm$. Direct rationalization gives

$$\operatorname{Im} F(\rho e^{i\theta}) = \frac{\rho^3 \sin \theta}{1 - xm} G(m, x), \quad G(m, x) = \int_0^1 \frac{m - u}{D(u)} d\omega(u). \quad (4.2)$$

At $m = 0$, $G < 0$. At $m = 1$ and $x < 1$, $G > 0$. Its derivative is

$$G_m(m, x) = \int_0^1 \frac{1 - xu}{D(u)^2} d\omega(u) > 0. \quad (4.3)$$

Therefore there is exactly one root $0 < m < 1$ at every $x < 1$.

At $x = 1$, an endpoint argument is essential. Put $\varepsilon = 1 - m$. Then $D(u) = (1 - u)^2 + \varepsilon u$. On the negative part $u > m \geq 1/2$,

$$\left| \frac{m - u}{D(u)} \right| \leq \frac{1}{u} \leq 2.$$

Its integral tends to zero, since $\omega((m, 1)) \rightarrow 0$. On any fixed interior interval below m , the integrand tends to $(1 - u)^{-1} > 0$. Thus $G(m, 1) > 0$ for m sufficiently close to one. This argument does not require $\int (1 - u)^{-1} d\omega$ to be finite; in the subcritical application it is infinite. Equation (4.3) now proves the unique root at $x = 1$ as well.

For actual angular parameters, the numerator $2c - \rho(1 + u)$ is negative when $c \leq \rho/2$ and positive when $c \geq \rho$, whenever these ranges occur. Inside this interval G increases with m , whereas m decreases with θ . This proves the global sign pattern, the strict range, and simplicity. Smooth dependence follows from the implicit function theorem, since the root stays away from $z = 1$ even when $\rho = 1$.

For the radial sign, set $d(u) = u^2 - (1 + m)u$. At a root of G , the probability measure proportional to $D(u)^{-1} d\omega(u)$ has mean m . Since

$$d(m) = -m, \quad d(u) - d(m) = (u - m)(u - 1),$$

subtracting the constant $d(m)/(1 - xm)$ in the expression for G_x gives

$$G_x = -\frac{1}{1 - xm} \int_0^1 \frac{(u - m)^2(1 - u)}{D(u)^2} d\omega(u) < 0.$$

It follows that

$$\frac{dm}{dx} = \frac{\int_0^1 \frac{(u - m)^2(1 - u)}{D(u)^2} d\omega(u)}{(1 - xm) \int_0^1 \frac{1 - xu}{D(u)^2} d\omega(u)} > 0. \quad (4.4)$$

Finally $\eta = (1 + m)/2$, so $\eta'(\rho) = \rho m'(\rho^2) > 0$. Strict positivity follows from the nondegenerate interior density. $\square$

Theorem 4.3 (Resolution of subcritical angular uniqueness). *For all $a, b > 0$ with $a + b \leq 1$ and $0 < \rho \leq 1$, the unique upper semicircle zero $\theta_{a,b}(\rho)$ satisfies (4.1). Its normalized displacement*

$$\eta_{a,b}(\rho) = \frac{\cos \theta_{a,b}(\rho)}{\rho}$$

increases strictly and lies in $(1/2, 1)$. Formula (4.4), with $\omega = \omega_{a,b}$, is an exact positive expression for the derivative.

Proof. Apply Lemma 4.2 to the finite positive measure in Corollary 3.3. $\square$

This proves the conjecture labeled `conj:subcritical` in [2], including unit radius and simplicity. The extension of normalized-radius monotonicity from the critical line to the entire subcritical triangle is an additional conclusion.

Corollary 4.4 (Angular uniqueness at every positive pair). *For all $a, b > 0$ and $0 < \rho \leq 1$, there is exactly one simple zero of $\text{Im } F_{a,b}(\rho e^{i\theta})$ in $0 < \theta < \pi$. The sign is positive before it and negative after it, and*

$$\arccos \rho < \theta_{a,b}(\rho) < \pi/2.$$

Proof. Only $w > 1$ remains. Let μ be the one-crossing finite measure in Proposition 2.3, and set

$$K_{\rho,c}(u) = \frac{1}{1 - 2\rho cu + \rho^2 u^2}, \quad J(\rho, c) = \int K_{\rho,c} d\mu.$$

The imaginary part equals $\rho \sin \theta J(\rho, c)$. An increasing test function integrates positively against a nonzero zero-mass measure whose negative support lies to the left of its positive support: subtract its value at the crossing. Thus $J(\rho, 0) < 0$ and $J(\rho, \rho) > 0$ for $\rho < 1$. At $\rho = 1$, the negative part is supported away from one and

$$\lim_{c \uparrow 1} J(1, c) = \int (1 - u)^{-2} d\mu(u) = \sum_{n \geq 1} n c_n > 0,$$

with $+\infty$ allowed. Separate the signs before applying monotone convergence or expanding the nonnegative polynomial partial sums.

For $c' > c$ the ratio $K_{\rho,c'}/K_{\rho,c}$ increases strictly on $(0, 1)$, since its derivative is

$$\frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0.$$

At a zero of J , apply the same ordered-sign test to $K_{\rho,c}\mu$, which again has zero mass. This proves uniqueness and the global sign pattern. At the zero,

$$J_c = \int K_{\rho,c}(u) \frac{2\rho u}{1 - 2\rho cu + \rho^2 u^2} d\mu(u) > 0,$$

because the second factor increases strictly. The angular derivative is $-\rho \sin^2 \theta J_c < 0$. This is the prior supercritical argument, now combined with the newly proved subcritical case. $\square$

Corollary 4.5 (Strict small-radius coefficient below the threshold). *For $w \leq 1$,*

$$\eta_{a,b}(\rho) = \frac{c_3}{2c_2} + \left(\frac{c_3 c_4}{c_2^2} - \frac{c_5}{2c_2} - \frac{c_3^3}{2c_2^3} \right) \rho^2 + O(\rho^4). \quad (4.5)$$

The coefficient of ρ^2 is strictly positive.

Proof. At $x = 0$, $m_0 = (c_3 - c_2)/c_2$ is the mean of U with law ω/c_2 . Formula (4.4) gives the coefficient $\frac{1}{2}\mathbb{E}[(U - m_0)^2(1 - U)] > 0$. The moments $\int u^j d\omega = c_{j+2} - c_{j+1}$ expand it into (4.5). Analytic implicit dependence near $x = 0$ justifies the remainder. The expansion itself is not new; the strict sign on the whole subcritical region is. $\square$

5 A fixed-weight Gaussian monotonicity theorem

Theorem 5.1 (Strict monotonicity along every fixed total weight). *Fix $w > 0$ and $0 < \rho \leq 1$. The function*

$$a \mapsto -\operatorname{Im} F_{a,w-a}(i\rho) \quad (5.1)$$

is strictly decreasing on $0 < a < w$.

Proof. In (2.6), put $x = e^{-s}$, $y = e^{-(s+t)}$ and $T = s + t$. For $X = \rho e^{-s}$ and $Y = \rho e^{-T}$, define

$$Q_\rho(s, T) = \frac{XY(X + Y)}{(1 + X^2)(1 + Y^2)}. \quad (5.2)$$

Then

$$\begin{aligned} -\operatorname{Im} F_{a,b}(i\rho) &= \frac{1}{\Gamma(a)\Gamma(b)} \int_0^\infty \int_0^\infty s^{a-1} t^{b-1} Q_\rho(s, s+t) dt ds \\ &= \frac{1}{\Gamma(w)} \int_0^\infty T^{w-1} \mathbb{E}[Q_\rho(TV, T)] dT, \quad V \sim \operatorname{Beta}(a, w-a). \end{aligned} \quad (5.3)$$

For fixed T , the function $s \mapsto Q_\rho(s, T)$ decreases strictly. Indeed X decreases with s , and

$$\frac{\partial}{\partial X} \frac{XY(X + Y)}{(1 + X^2)(1 + Y^2)} = \frac{Y\{2X + Y(1 - X^2)\}}{(1 + X^2)^2(1 + Y^2)} > 0, \quad (5.4)$$

where $0 < X \leq \rho \leq 1$.

For $a_2 > a_1$, the ratio of the two beta densities is a positive constant times $(v/(1-v))^{a_2-a_1}$, which increases strictly. The expectation of any strictly decreasing function therefore decreases strictly. One direct proof differentiates in a :

$$\frac{\partial}{\partial a} \mathbb{E}[q(V)] = \operatorname{Cov}\left(q(V), \log \frac{V}{1-V}\right) < 0. \quad (5.5)$$

The sign follows by writing covariance as half the expectation of $(q(V) - q(V'))(L(V) - L(V'))$ for independent V, V' ; the factors have opposite signs off the diagonal. Apply this with $q(v) = Q_\rho(Tv, T)$. Since $Q_\rho(Tv, T) \leq 2\rho^3 e^{-T}$, the T integral is dominated. Beta logarithmic moments are locally uniformly finite for $0 < a < w$, so differentiation and integration can be interchanged as well. Strictness survives integration over every $T > 0$. $\square$

5.1 Explicit endpoint identities and inequalities

Write $\beta(s)$ for the analytically continued Dirichlet beta function, and $\eta_D(s) = (1 - 2^{1-s})\zeta(s)$ for the Dirichlet eta function, with its removable value at $s = 1$. This notation is distinct from the normalized angular displacement $\eta_{a,b}(\rho)$.

Corollary 5.2 (Sharp fixed-weight Gaussian bounds). *For $0 < a < w$,*

$$\beta(w) - \beta(w-1) < -g_{a,w-a} < \frac{\beta(w) + 2^{-w}\eta_D(w)}{2}. \quad (5.6)$$

Both endpoints are sharp limits, attained as $a \uparrow w$ and $a \downarrow 0$, respectively, but neither is attained at an interior parameter. For every $w > 0$ and every $a \in (0, w)$,

$$-2g_{a,w-a} < \frac{1 + \sqrt{2}}{2}. \quad (5.7)$$

Proof. As $a \downarrow 0$, the beta variable in (5.3) concentrates at zero; as $a \uparrow w$, it concentrates at one. The preceding domination proves convergence of the expectations and integrals. The limiting analytic functions are

$$F_{0,w}(z) = \frac{z}{1-z} \operatorname{Li}_w(z), \quad F_{w,0}(z) = \operatorname{Li}_{w-1}(z) - \operatorname{Li}_w(z). \quad (5.8)$$

These identities follow first by coefficient summation in the disk, and their integral continuations give the endpoint values at $z = i$. At that point,

$$\text{Li}_w(i) = -2^{-w} \eta_D(w) + i\beta(w).$$

This yields (5.6). When $w \leq 1$, the occurrence of $\beta(w - 1)$ is an analytic value, not the sum of a convergent ordinary Dirichlet series.

Twice the upper limit in (5.6) also equals

$$\frac{1}{\Gamma(w)} \int_0^\infty T^{w-1} e^{-T} \frac{1 + e^{-T}}{1 + e^{-2T}} dT.$$

The scalar function $(1 + y)/(1 + y^2)$ on $0 < y < 1$ has maximum $(1 + \sqrt{2})/2$, at $y = \sqrt{2} - 1$. Since the gamma measure is not concentrated there, its expectation is strictly smaller. $\square$

Remark 5.3 (Why the restriction on the Gaussian radius matters). The derivative in (5.4) is uniformly positive in the present argument because $X \leq 1$. For radii above one, $Y(1 - X^2)$ can have the other sign. The theorem does not assert monotonicity for all $\rho > 1$, and the proof cannot be extended there by dropping a hypothesis.

6 Euler acceleration without a finite signed measure

6.1 A bounded order-one transport functional

Let

$$d\sigma_a(v) = k_a(-\log v) dv, \quad d\tau_b(u) = \frac{k_b(-\log u)}{1 - u} du \quad (0 < u, v < 1). \quad (6.1)$$

The measure σ_a is a probability measure, while τ_b may have infinite mass. Nevertheless $\int (1 - u) d\tau_b(u) = 1$.

Theorem 6.1 (All-order positive transport). *For all $a, b > 0$, the functional*

$$\mathcal{L}_{a,b}(\varphi) = \int_0^1 \int_0^1 (\varphi(v) - \varphi(uv)) d\sigma_a(v) d\tau_b(u) \quad (6.2)$$

is well-defined for $\varphi \in C^1[0, 1]$, and

$$|\mathcal{L}_{a,b}(\varphi)| \leq 2^{-a} \|\varphi'\|_\infty. \quad (6.3)$$

The constant is exact. Its polynomial moments are $\mathcal{L}_{a,b}(u^{n-1}) = c_n$, and

$$F_{a,b}(z) = \mathcal{L}_{a,b} \left(u \mapsto \frac{z}{1 - zu} \right) \quad (z \in \Omega). \quad (6.4)$$

When $a + b < 1$, this functional cannot satisfy a bound by $C\|\varphi\|_\infty$: its distributional order is exactly one, not zero.

Proof. The mean-value estimate is $|\varphi(v) - \varphi(uv)| \leq v(1 - u)\|\varphi'\|_\infty$. Its integral is $2^{-a}\|\varphi'\|_\infty$; equality occurs for $\varphi(u) = u$. For a monomial,

$$\mathcal{L}(u^{n-1}) = \left(\int v^{n-1} d\sigma_a(v) \right) \left(\int \frac{1 - u^{n-1}}{1 - u} k_b(-\log u) du \right) = n^{-a} H_{n-1}^{(b)}.$$

The power series of $z/(1-zu)$ converges in $C^1[0, 1]$ for $|z| < 1$, so the moment equality proves (6.4) there. On compact slit sets, the same test functions and their derivatives are uniformly bounded; hence their images under $\mathcal{L}$ are holomorphic and continuation proves the formula everywhere. For $w < 1$, the test polynomials u^{n-1} have supremum norm one but $\mathcal{L}(u^{n-1}) = c_n \rightarrow \infty$, excluding any order-zero bound. $\square$

6.2 Strict Euler signs and explicit remainders

For an integer stride $h \geq 1$, put

$$d_k^{(h)} = \sum_{n=0}^k (-1)^n \binom{k}{n} c_{hn+1}. \quad (6.5)$$

Theorem 6.2 (All-positive-order Euler differences). *For every $a, b > 0$ and $h \geq 1$,*

$$d_0^{(h)} = 0, \quad 0 < -d_k^{(h)} < H_h^{(b)} \leq h \quad (k \geq 1). \quad (6.6)$$

More precisely,

$$-d_k^{(h)} = \iint \left[(1 - u^h v^h)^k - (1 - v^h)^k \right] d\sigma_a(v) d\tau_b(u). \quad (6.7)$$

For the Gaussian stride $h = 2$, let

$$E_N = \sum_{k=0}^{N-1} \frac{d_k^{(2)}}{2^{k+1}} \quad (N \geq 1). \quad (6.8)$$

Then E_N decreases strictly to the analytic value $g_{a,b}$, and

$$E_N - H_2^{(b)} 2^{-N} < g_{a,b} < E_N. \quad (6.9)$$

The exact positive tail identity is

$$2^N (E_N - g_{a,b}) = \iint [q_N(uv) - q_N(v)] d\sigma_a(v) d\tau_b(u), \quad q_N(v) = \frac{(1 - v^2)^N}{1 + v^2}. \quad (6.10)$$

Proof. The binomial theorem inside the finite sum gives (6.7). It is strictly positive. For $U = u^h$, $V = v^h$, convexity of $f(t) = (1 - t)^k$ gives

$$0 < f(UV) - f(V) \leq (1 - U)\{1 - f(V)\} < 1 - U.$$

Integrate and use $\int (1 - u^h) d\tau_b(u) = H_h^{(b)}$. This also justifies every integral despite a possibly infinite mass of τ_b .

The Euler series is absolutely convergent by the uniform bound on $d_k^{(2)}$. Tonelli applies to the positive integrands for $-d_k$. Summing them gives

$$\sum_{k \geq 0} \frac{d_k^{(2)}}{2^{k+1}} = \iint \left(\frac{1}{1 + v^2} - \frac{1}{1 + u^2 v^2} \right) d\sigma_a(v) d\tau_b(u) = g_{a,b},$$

where the last equality is (6.4) at $z = i$. Summing the remaining geometric series proves (6.10). Since q_N decreases strictly, its integrand is positive. Finally sum the strict coefficient bound over $k \geq N$ to get (6.9). $\square$

The estimate $H_2^{(b)} < 2$ is a convenient rational certificate bound at arbitrary positive orders. It is not claimed to be the optimal all-order Gaussian error constant. The preceding constant one for $a \geq 1$ remains sharper on its stated domain and is not superseded by a weaker estimate. The coefficient bounds themselves have sharp uniform constants: for $h = 1$, let $a \downarrow 0$, $b \rightarrow \infty$ to obtain $d_k^{(1)} \rightarrow -1$ for fixed $k \geq 1$; for $h = 2$, already $-d_1^{(2)} = H_2^{(b)}/3^a \rightarrow 2$ as $a, b \downarrow 0$.

Corollary 6.3 (Independent affine strides). *Let $a, b > 0$, let r be a positive integer, and let $\lambda > 0$. For $f(n) = H_{rn}^{(b)}/(\lambda n + 1)^a$, define $d_k = \sum_{n=0}^k (-1)^n \binom{k}{n} f(n)$. Then $d_0 = 0$ and $0 < -d_k < H_r^{(b)}$ for every $k \geq 1$. The Euler sums $E_N(f) = \sum_{k < N} d_k/2^{k+1}$ decrease strictly to the Abel value $g(f) = \lim_{t \uparrow 1} \sum_{n \geq 0} (-t)^n f(n)$, with*

$$E_N(f) - H_r^{(b)} 2^{-N} < g(f) < E_N(f).$$

Thus the positivity and Euler remainder extend to independent strides without requiring $a \geq 1$ or $a + b \geq 1$.

Proof. The coefficient is $\iint v^{\lambda n} (1 - u^{rn}) d\sigma_a(v) d\tau_b(u)$. The preceding convexity argument applies with $U = u^r$ and $V = v^\lambda$, and $\int (1 - u^r) d\tau_b = H_r^{(b)}$. For $0 < t < 1$, geometric summation gives the integrand $(1 + tv^\lambda)^{-1} - (1 + tu^r v^\lambda)^{-1}$. Its absolute value is at most $1 - u^r$, so dominated convergence identifies the Abel limit. The same positive Euler-tail summation proves the claimed enclosure. This proves an evaluator theorem, not an assertion about an affine one-crossing finite signed kernel. $\square$

Proposition 6.4 (A complex Euler identity and a sharp negative-axis tail). *For $\operatorname{Re} z < 1/2$, put $q = z/(z - 1)$. Then*

$$F_{a,b}(z) = \frac{z}{1 - z} \sum_{k \geq 0} d_k^{(1)} q^k, \quad (6.11)$$

and truncation before $k = N$ has error at most

$$\frac{|z|}{|1 - z|} \frac{|q|^N}{1 - |q|}. \quad (6.12)$$

For $z = -r$, $r > 0$, the truncated values increase to $F_{a,b}(-r) > 0$, and the error is strictly less than

$$r \left(\frac{r}{1 + r} \right)^N \quad (N \geq 1). \quad (6.13)$$

The coefficient one multiplying this last bound is sharp uniformly over positive a, b at each fixed r, N .

Proof. Set $A(t) = \sum_{n \geq 0} c_{n+1} t^n = F(t)/t$. The binomial transform gives

$$\sum_{k \geq 0} d_k^{(1)} q^k = \frac{1}{1 - q} A\left(\frac{-q}{1 - q}\right)$$

initially near zero. The left side is analytic for $|q| < 1$ by $|d_k^{(1)}| < 1$. Substitution $q = z/(z - 1)$ maps this disk to $\operatorname{Re} z < 1/2$ and analytic continuation proves the identity. The geometric tail proves (6.12). At $z = -r$, $q = r/(1 + r)$ and the prefactor is $-q$; since all the nonzero d_k are negative, monotonicity and the strict bound follow. For fixed k , $d_k \rightarrow -1$ as $a \downarrow 0$, $b \rightarrow \infty$. Dominated convergence in the geometrically weighted tail yields exactly $r q^N$, proving sharpness. $\square$

7 The sharp critical Euler constant

Theorem 7.1 (Exact fixed-pair constant on the closed subcritical region). *For $a, b > 0$ with $a + b \leq 1$, the sequence $2^N(E_N - g_{a,b})$, $N \geq 1$, decreases strictly to zero. In fact*

$$2^N(E_N - g_{a,b}) = \int_0^1 \frac{(1 - u^2)^N}{1 + u^2} d\nu_{a,b}(u). \quad (7.1)$$

The least constant C such that $E_N - g_{a,b} \leq C2^{-N}$ for every $N \geq 1$ is

$$C(a, b) = -2g_{a,b}. \quad (7.2)$$

For strict inequalities, the infimum is the same, but is not attained at a fixed pair because equality occurs at $N = 1$.

Proof. For $k \geq 1$, the finite compensated moment formula gives

$$d_k^{(2)} = - \int_0^1 (1 - u^2)^k d\nu_{a,b}(u).$$

Its integral is finite: near one, $(1 - u^2)^k = O(1 - u)$, and $(1 - u)\nu$ is finite. Summing the positive tail proves (7.1). For $N \geq 1$ its integrand is dominated by $q_1(u) \leq 2(1 - u)$ and decreases strictly to zero for $0 < u < 1$. Dominated convergence proves the asserted decrease and limit even when ν has infinite mass. The largest scaled error is at $N = 1$, where $E_1 = 0$, so it equals $-2g_{a,b}$. $\square$

Theorem 7.2 (Resolution of the sharp critical constant conjecture). *The function $a \mapsto -2g_{a,1-a}$ decreases strictly on $(0, 1)$, with*

$$\lim_{a \downarrow 0} (-2g_{a,1-a}) = \frac{\pi}{4} + \frac{\log 2}{2}, \quad \lim_{a \uparrow 1} (-2g_{a,1-a}) = \frac{\pi}{2} - 1. \quad (7.3)$$

The sharp constant uniform over $0 < a < 1$, $b = 1 - a$, and $N \geq 1$ is

$$C_{\text{critical}} = \frac{\pi}{4} + \frac{\log 2}{2}. \quad (7.4)$$

In particular,

$$E_N - C_{\text{critical}}2^{-N} < g_{a,1-a} < E_N \quad (0 < a < 1, N \geq 1). \quad (7.5)$$

No smaller uniform constant works, even if the inequalities are allowed to be non-strict.

Proof. The strict monotonicity is Theorem 5.1 at $w = 1$, $\rho = 1$. Corollary 5.2, using $\beta(1) = \pi/4$, $\beta(0) = 1/2$, and $\eta_D(1) = \log 2$, gives the endpoint values. The fixed-pair constant is Theorem 7.1. Its supremum is therefore exactly (7.4). It is not attained by an interior pair, which proves the strict uniform bound. For any smaller candidate, take a sufficiently close to zero and $N = 1$ to contradict it. $\square$

This proves both parts of the conjecture labeled `conj:constant` in [2]: strict parameter monotonicity, not merely an upper bound, and the matching optimal uniform constant. Numerically, $C_{\text{critical}} \approx 1.1319717536774209643$. Theorem 7.1 and (5.7) also give a uniform, not asserted optimal, constant $(1 + \sqrt{2})/2$ on the whole closed subcritical triangle.

8 Strict polylogarithms at arbitrary depth

Let $d \geq 1$, $\mathbf{s} = (s_1, \dots, s_d)$ with $s_j > 0$, and $w = \sum s_j$. Define

$$F_{\mathbf{s}}(z) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{s_1} \dots n_d^{s_d}} = \sum_{n \geq d} c_n(\mathbf{s}) z^n. \quad (8.1)$$

The notation $c_n(\mathbf{s})$ in this section generalizes, and does not alter, the depth-two convention.

Theorem 8.1 (Ordered-simplex positive kernel). *For $z \in \Omega$,*

$$F_{\mathbf{s}}(z) = z^d \int_{1 > x_1 > \dots > x_d > 0} \frac{W_{\mathbf{s}}(\mathbf{x})}{\prod_{j=1}^d (1 - zx_j)} d\mathbf{x}, \quad (8.2)$$

where $x_0 = 1$ and

$$W_{\mathbf{s}}(\mathbf{x}) = \prod_{j=1}^d \frac{\log(x_{j-1}/x_j)^{s_j-1}}{\Gamma(s_j)}. \quad (8.3)$$

The weight is positive, integrable, and has mass

$$M_{\mathbf{s}} = \prod_{j=1}^d (d-j+1)^{-s_j} = c_d(\mathbf{s}). \quad (8.4)$$

Proof. Insert $n_j^{-s_j} = \int_0^\infty e^{-n_j t_j} k_{s_j}(t_j) dt_j$ and write the positive integer gaps $k_j = n_j - n_{j+1}$, with $n_{d+1} = 0$. Set $T_j = t_1 + \dots + t_j$. Then $\sum n_j t_j = \sum k_j T_j$ and $n_1 = \sum k_j$. The sums over the k_j are independent geometric series, giving

$$\int_{\mathbb{R}_+^d} \prod_{j=1}^d k_{s_j}(t_j) \prod_{j=1}^d \frac{ze^{-T_j}}{1 - ze^{-T_j}} d\mathbf{t}.$$

All exchanges are justified by absolute convergence for $|z| < 1$. Now set $x_j = e^{-T_j}$. The Jacobian is $\prod x_j$, exactly canceling the product in the numerators. This yields (8.2). For its mass, $\prod x_j = e^{-\sum_{i=1}^d (d-i+1)t_i}$, so independent gamma integrals give (8.4). Finite mass and denominator separation on compact slit sets prove holomorphic continuation as in Lemma 2.2. $\square$

8.1 A positive generalized Stieltjes measure

Let $\Delta_{d-1} = \{p_j \geq 0 : \sum p_j = 1\}$, with uniform Dirichlet probability measure (density $\Gamma(d)$ in its usual $d-1$ coordinates). For $d=1$ this is the point mass at $p_1 = 1$.

Theorem 8.2 (Depth- d generalized Stieltjes representation). *There is a finite positive measure $\mu_{\mathbf{s}}$ on $(0, 1)$, with mass $M_{\mathbf{s}}$ and positive mass on every nonempty interior interval, such that*

$$\frac{F_{\mathbf{s}}(z)}{z^d} = \int_0^1 (1 - zu)^{-d} d\mu_{\mathbf{s}}(u) \quad (z \in \Omega). \quad (8.5)$$

Explicitly, $\mu_{\mathbf{s}}$ is the pushforward of $W_{\mathbf{s}}(\mathbf{x}) d\mathbf{x}$ times the uniform Dirichlet probability law under $u = \sum_{j=1}^d p_j x_j$.

Proof. For $\operatorname{Re} A_j > 0$, independent Laplace integrals followed by $r_j = tp_j$ give the classical parameter identity

$$\prod_{j=1}^d A_j^{-1} = \Gamma(d) \int_{\Delta_{d-1}} \left(\sum_{j=1}^d p_j A_j \right)^{-d} d\mathbf{p}. \quad (8.6)$$

Apply it with $A_j = 1 - zx_j$ first for $\operatorname{Re} z < 1$. Absolute integrability against the finite positive weight justifies the iterated integral. The proposed pushforward then proves (8.5); analytic continuation gives the whole slit plane. Since every x_j lies in $(0, 1)$, so does u . For any open interval $I \subset (0, 1)$, choose all strictly ordered x_j in I . That set has positive W -mass and its every convex combination lies in I . Hence $\mu_s(I) > 0$. $\square$

The notion of exact generalized Stieltjes order is standard; see Karp and Prilepkina [5]. We prove the needed minimality from a decay obstruction rather than invoking an exact-order criterion. The recent star-polylogarithm representation of Li [6] concerns weakly decreasing indices and different normalizations. It is not a contradiction to the exact order found here for the strictly nested function divided by z^d .

8.2 A negative-axis asymptotic that forces exact order

Theorem 8.3 (Universal logarithmic leading term). *For every positive index vector $\mathbf{s}$,*

$$(-1)^d F_{\mathbf{s}}(-x) \sim \frac{(\log x)^w}{\Gamma(w+1)} \quad (x \rightarrow +\infty). \quad (8.7)$$

The leading coefficient depends only on the total weight, not on the individual positive orders.

Proof. Put $x = e^L$ and $\ell(y) = (1 + e^{-y})^{-1}$. The Mellin representation in the proof of Theorem 8.1 gives the exact positive formula

$$(-1)^d F_{\mathbf{s}}(-e^L) = \frac{1}{\prod_j \Gamma(s_j)} \int_{\mathbb{R}_+^d} \prod_j t_j^{s_j-1} \prod_{j=1}^d \ell(L - T_j) dt. \quad (8.8)$$

Rescale $t_j = Lv_j$ for $L \geq 1$. Except on a set of measure zero, the product of logistic factors tends to the indicator of $\sum v_j < 1$, because the cumulative sums increase with j . On $\sum v_j \leq 1$ the product is bounded by one. On $r = \sum v_j > 1$, the last factor is at most $e^{-L(r-1)} \leq e^{-(r-1)}$, while all others are at most one. These bounds, multiplied by $\prod_j v_j^{s_j-1}$, are integrable: radial integration reduces the outside bound to a constant times $\int_1^\infty r^{w-1} e^{-(r-1)} dr$. Dominated convergence now gives

$$\lim_{L \rightarrow \infty} \frac{(-1)^d F_{\mathbf{s}}(-e^L)}{L^w} = \frac{1}{\prod_j \Gamma(s_j)} \int_{v_j > 0, \sum v_j < 1} \prod_j v_j^{s_j-1} d\mathbf{v} = \frac{1}{\Gamma(w+1)}.$$

$\square$

Definition 8.4. A generalized Stieltjes function of order $\alpha > 0$ is a function on $x > 0$ representable as

$$f(x) = c + \int_{[0, \infty)} (x+t)^{-\alpha} d\lambda(t), \quad c \geq 0, \quad \lambda \geq 0,$$

with $\int (1+t)^{-\alpha} d\lambda(t) < \infty$. Its exact order is the infimum of the possible positive orders.

Theorem 8.5 (Exact generalized Stieltjes order equals strict depth). *The function*

$$H_s(x) = \frac{F_s(-x)}{(-x)^d} \quad (x > 0) \quad (8.9)$$

has exact generalized Stieltjes order d . It has no positive representation of any order $0 < \alpha < d$, even when the representing measure is allowed on the entire half-line rather than a compact interval.

Proof. Formula (8.5) represents H_s as $\int (1+xu)^{-d} d\mu_s(u)$. With $t = 1/u$ and the weighted pushforward $d\lambda(t) = u^{-d} d\mu_s(u)$, this is a generalized Stieltjes representation of order d . The required integrability follows from $\int (1+u)^{-d} d\mu_s < \infty$.

By Theorem 8.3,

$$H_s(x) \sim \frac{(\log x)^w}{\Gamma(w+1)x^d}. \quad (8.10)$$

Suppose it had an order- α representation for $\alpha < d$. Since $H_s(x) \rightarrow 0$, the constant term is zero. The nonzero positive measure must assign a finite positive mass m to some $[0, R]$; local finiteness follows from the defining weighted integrability. Then $H_s(x) \geq m(x+R)^{-\alpha}$. But (8.10) implies $x^\alpha H_s(x) \rightarrow 0$, a contradiction. $\square$

9 The exact coefficient-normalization threshold

For $\alpha > 0$, let $(\alpha)_n = \Gamma(\alpha+n)/\Gamma(\alpha)$ and set

$$M_n^{(\alpha)} = \frac{n!}{(\alpha)_n} c_{n+d}(\mathbf{s}), \quad n \geq 0. \quad (9.1)$$

Theorem 9.1 (Positive Hausdorff normalization if and only if $\alpha \geq d$). *The sequence $(M_n^{(\alpha)})_{n \geq 0}$ is a positive Hausdorff moment sequence on $[0, 1]$ exactly when $\alpha \geq d$. At $\alpha = d$, its measure is μ_s . At $\alpha > d$, a measure is obtained by replacing U by BU , where U has normalized law μ_s/M_s and $B \sim \text{Beta}(d, \alpha - d)$ is independent, then multiplying the law by M_s . For every $\alpha \geq d$, all finite alternating differences are strictly positive and all finite principal Hankel matrices are positive definite.*

Proof. Expand (8.5) inside the disk:

$$c_{n+d}(\mathbf{s}) = \frac{(d)_n}{n!} \int_0^1 u^n d\mu_s(u).$$

This proves the assertion at $\alpha = d$. For $\alpha > d$, $\mathbb{E}[B^n] = (d)_n/(\alpha)_n$, giving the product-measure construction. The measure assigns positive mass to every interior interval. Therefore

$$\sum_{j=0}^k (-1)^j \binom{k}{j} M_{n+j}^{(\alpha)} = \int u^n (1-u)^k d\mu^{(\alpha)}(u) > 0,$$

and the quadratic-form argument for a nonzero polynomial gives strict positive definiteness of every finite Hankel matrix.

Conversely, suppose $\alpha < d$ and these numbers were moments of a positive finite measure λ on $[0, 1]$. Summing its moments would give

$$\frac{F_s(z)}{z^d} = \int_0^1 (1-zu)^{-\alpha} d\lambda(u)$$

in the disk, and hence on the slit plane. A possible atom at zero is a nonnegative constant; the remainder, after $t = 1/u$, is a generalized Stieltjes function of order α . This contradicts Theorem 8.5. $\square$

Example 9.2 (A finite witness against the unnormalized order-one claim). For $\mathbf{s} = (1, 1)$, depth $d = 2$, and $\alpha = 1$,

$$M_0^{(1)} = \frac{1}{2}, \quad M_1^{(1)} = \frac{1}{2}, \quad M_2^{(1)} = \frac{11}{24}.$$

Their second alternating difference is $M_0^{(1)} - 2M_1^{(1)} + M_2^{(1)} = -1/24 < 0$. In contrast, the correctly normalized coefficients $c_{n+2}/(n+1)$ are positive Hausdorff moments at every positive pair of real indices. An omitted normalization is thus a checkable error, not a harmless convention.

Corollary 9.3 (Integral and differential normalization identities). *Put $A_{\mathbf{s}}(z) = F_{\mathbf{s}}(z)/z^d$, with its removable value at zero, and*

$$\mathcal{M}_{\mathbf{s}}(z) = \sum_{n \geq 0} \frac{c_{n+d}(\mathbf{s})}{\binom{n+d-1}{d-1}} z^n = \int_0^1 \frac{d\mu_{\mathbf{s}}(u)}{1-zu}.$$

For $d \geq 2$ and $z \in \Omega$,

$$\mathcal{M}_{\mathbf{s}}(z) = (d-1) \int_0^1 (1-t)^{d-2} A_{\mathbf{s}}(tz) dt, \quad (9.2)$$

$$A_{\mathbf{s}}(z) = \frac{1}{(d-1)!} \frac{d^{d-1}}{dz^{d-1}} [z^{d-1} \mathcal{M}_{\mathbf{s}}(z)]. \quad (9.3)$$

Proof. The beta integral gives $(d-1) \int_0^1 t^n (1-t)^{d-2} dt = 1/\binom{n+d-1}{d-1}$. This proves (9.2) by coefficient comparison. Differentiation multiplies the n th coefficient by the reciprocal factor, proving (9.3). Both sides are holomorphic on Ω : for a compact slit set, its products with $t \in [0, 1]$ form a compact subset of the same slit plane. Analytic continuation therefore establishes the stated domain. $\square$

Remark 9.4 (Analytic order is not arithmetic depth). The exact order d just proved is a positivity and decay invariant of this analytic function with its specified normalization. It does not prove that a special value cannot be reduced to lower-depth periods, that numerical constants are independent, or that a motivic depth filtration has a particular rank. These are distinct questions.

10 Reproducible exact certificates and independent tests

10.1 Fractional powers without floating-point assumptions

The standard-library script `code/certify.py` uses integer and rational arithmetic only. For $p/q > 0$ and integer $n \geq 1$, put $B = 10^{80}$ and compute

$$r = \left\lfloor (n^p B^q)^{1/q} \right\rfloor.$$

Before using the result, it verifies both integer inequalities

$$r^q \leq n^p B^q < (r+1)^q. \quad (10.1)$$

Except in the exact-root case, these imply $B/(r+1) < n^{-p/q} < B/r$. The exact-root case is handled separately. Endpoints are rounded outward onto $B^{-1}\mathbb{Z}$ using integer division. Positive harmonic

sums and products then enclose c_n . No floating-point root or transcendental sign is accepted as a proof.

The finite Euler sum is formed using the exact weights

$$E_N = 2^{-N} \sum_{n=0}^{N-1} (-1)^n c_{2n+1} \sum_{j=n+1}^N \binom{N}{j}. \quad (10.2)$$

Substitution of (6.5) proves this identity by the finite relation $\sum_{k=n}^{N-1} 2^{N-k-1} \binom{k}{n} = \sum_{j=n+1}^N \binom{N}{j}$, which follows from Pascal's recurrence. We use $N = 192$ and the proved rational tail constant 2 from Theorem 6.2. The finite centers at fractional orders are algebraic numbers enclosed by rational intervals; they are not mislabeled as rational exact values.

a	b	$g_{a,b}$ (displayed approximation only)
1/10	1/10	-0.490480773154378134692711376810
1/4	1/4	-0.464611276461282330148775822252
2/5	1/5	-0.424200725534215249048994001121
1/3	1/3	-0.446085363511268827586991313068
1/10	9/10	-0.529761932285771263094629365374
1/2	1/2	-0.403936987717013579833068109579
9/10	1/10	-0.306037812999015482508621228917
1	1	-0.272198261287950266312586112280
2	3	-0.093247950371174729821546998597

The exact JSON records both rational endpoints and outward decimal summaries. Every interval has width below 3.187×10^{-58} . For a shorter readable certified example,

$$\begin{aligned} -0.464611276461282330148775822252 &< g_{1/4,1/4} \\ &< -0.464611276461282330148775822251. \end{aligned}$$

This lies strictly below the old finite-measure threshold $a + b = 1$. It is a value enclosure, not a conjectured closed form.

10.2 Angular brackets with rational Chebyshev evaluation

For rational $0 < \rho < 1$ and $-1 < c < 1$, the exact identity is

$$\frac{\operatorname{Im} F_{a,b}(\rho e^{i\theta})}{\sin \theta} = \sum_{n \geq 1} c_n \rho^n U_{n-1}(c), \quad c = \cos \theta. \quad (10.3)$$

Use $c_n \leq n$ and $|U_{n-1}(c)| \leq n$. For $j = N + 1$, an explicit absolute tail bound is

$$\rho^j \left\{ \frac{j^2}{1-\rho} + \frac{2j\rho}{(1-\rho)^2} + \frac{\rho(1+\rho)}{(1-\rho)^3} \right\}. \quad (10.4)$$

This is the exact sum of $\sum_{n \geq j} n^2 \rho^n$.

For $c = P/Q$, the program computes integer numerators $U_k(P/Q) = V_k/Q^k$, where $V_0 = 1$, $V_1 = 2P$, and $V_k = 2PV_{k-1} - Q^2V_{k-2}$. Each rational summand is rounded outward separately. This avoids unstable interval recursion in the Chebyshev polynomial itself. The verifier uses $N = 400$, checks a strictly negative enclosure at the lower cosine endpoint and a strictly positive one at the upper endpoint, and checks that both lie between $\rho/2$ and ρ . The analytic theorem, not the finite root search, supplies uniqueness.

a	b	$\eta(1/4)$	$\eta(1/2)$	$\eta(3/4)$
1/10	1/10	0.9290066544	0.9320671593	0.9391522820
1/4	1/4	0.8330769037	0.8376577074	0.8478255853
2/5	1/5	0.7966236703	0.8012312684	0.8112578998
1/3	1/3	0.7848252542	0.7893512974	0.7991218117

These are numerical summaries. The twelve accepted rational cosine brackets each have width 5×10^{-12} . No trigonometric rounding is involved in their acceptance. No unit-radius bracket is claimed; unit radius is covered by the endpoint proof in Lemma 4.2.

10.3 Recorded evidence and replay

The exact verifier checks 3,915 integer-root inequalities, all nine Euler enclosures, and all twelve angular brackets. At integer orders it also verifies 728 strictly positive finite differences and 96 strictly positive Hankel determinants for eight index vectors through depth four, together with the exact counterexample $-1/24$ above. A further 288 exact coefficient checks cover independent affine strides at integer orders, including the noninteger denominator step $1/2$. The separate SymPy script checks seven rational identities used in the proofs, including the root derivative and the beta-kernel derivative. These finite instances are tests, not substitutes for the all-depth arguments.

Independent generalized Gauss–Laguerre quadrature uses 2^{-a} times a product of gamma probability laws after $x = e^{-S/2}$, $y = xe^{-T}$. Its 128-node comparison against the Euler centers has largest observed absolute discrepancy 3.109×10^{-15} in the recorded run. This is a floating-point diagnostic, not a certified quadrature error. Six fixed-weight grids with nine points each show strict decrease, again only as diagnostics.

From the package root, the exact replay is

```
python code/certify.py --check
```

It requires only the Python standard library. Optional diagnostics and symbolic tests are reproduced by

```
python code/diagnostics.py
python code/symbolic_checks.py
```

The first command regenerates rational bracket proposals; running `certify.py` afterward independently accepts or rejects them. For the frozen replay, do not regenerate proposals first. The file manifest checks byte integrity, not mathematical truth.

11 Audit conclusions and integration guidance

11.1 What is corrected, and what is only strengthened

No contradiction to the *stated* finite signed threshold or the previous critical endpoint theorem was found. Those results remain valid. The subcritical construction is a different representation, not a repair of a false finite-measure theorem. The precise status changes are that the prior angular-uniqueness and sharp-critical-constant conjectures now have proofs, and the separate subcritical slit-plane question has a positive answer.

There are four concrete hazards to correct in any integration or extrapolation of earlier material.

Infinite mass must remain compensated. Below the threshold, splitting the new positive integral into separate masses is invalid. Use (3.2) or the finite measure ω , never a fictitious finite signed measure. The threshold $a + b \geq 1$ and the first-difference threshold $a + b \leq 1$ are complementary, not competing claims about the same measure.

Boundary values are not ordinarily convergent sums. For $w < 1$ the coefficients grow, and on $w = 1$ they approach a nonzero constant. The new Euler theorem evaluates analytic/Abel values in these regimes. It must not be described as acceleration of an ordinarily convergent boundary series there.

The singular angular endpoint needs its own proof. At $\rho = 1$ below threshold, the limiting integral $\int (1-u)^{-1} d\omega$ is infinite. An argument using dominated convergence against that expression would be invalid. The negative-tail bound by $2\omega((m, 1))$ in Lemma 4.2 is the required replacement and also proves existence, not just uniqueness conditional on a zero being present.

Stieltjes normalizations and depth conventions matter. For strict depth d , the order-one moment sequence is $c_{n+d}/\binom{n+d-1}{d-1}$, not generally c_{n+d} . The exact counterexample in Section 9 shows that dropping the denominator can falsify even a three-term positivity test. A star polylogarithm theorem must not be transferred to strict nesting without checking this change. Exact analytic Stieltjes order is not an arithmetic non-reducibility theorem for its special values.

11.2 Suggested insertion points

The complete report can be placed at `Analysis/Polylogarithms/docs/reports/compensated-polylogarithms/`. The supplied insertion fragment uses the prefix `compensated:`. It summarizes the new theorems and points back to the full proofs. The current main file inputs `chapters/05-certified-computation.tex`; that chapter and the signed-kernel discussion are natural insertion points for the compensated and Euler results. The zero-geometry chapter and final research-programme chapter should cross-reference the two resolved conjectures. The all-depth Stieltjes theorem fits with the foundational iterated-integral material.

Integration should preserve the historical record: label the original conjectures as resolved by this continuation, do not rewrite them as if they had already been proved in the earlier report. Keep the supercritical normalized-radius conjecture, S_6 , and numerical period independence questions explicitly open. A stale or differently organized chapter path in an earlier integration note should be mapped to the current main-file inputs rather than copied blindly.

12 Further research: precise next questions

12.1 The remaining normalized-radius phase diagram

We have proved strict increase for $a + b \leq 1$. The earlier conjecture for integer $a \geq 2$ and finite $b > 0$, and the sign transition at $a = 1$, are not settled by this argument. A first objective is a sharp description of the set of (a, b) for which the radial sign changes as ρ varies. The small-radius coefficient in (4.5) is an exact local test, but a sign for that coefficient does not imply a global sign. The positive variance mechanism here works because the factor is $1 - z$, equivalently because

$d(u) - d(m) = (u - m)(u - 1)$; an interior crossing factor $1 - rz$ need not have the same sign structure.

12.2 Optimal Euler constants beyond the critical line

The exact fixed-pair constant is now known on $a + b \leq 1$. A sharper uniform constant on the whole subcritical triangle is reduced to understanding the envelope

$$C_*(w) = \beta(w) + 2^{-w} \eta_D(w), \quad 0 < w \leq 1,$$

since the fixed-weight supremum occurs as $a \downarrow 0$. The bound $(1 + \sqrt{2})/2$ is explicit but is not asserted optimal. On the full positive quadrant, the all-order Euler bound $H_2^{(b)}$ is valid, but scaled-error monotonicity in N is not proved here. Finding the actual universal Gaussian Euler constant remains a separate optimization problem. Uniform error asymptotics near $w = 1$ would also clarify how infinite compensation crosses over to a finite critical measure.

12.3 Higher-depth angular geometry

The exact Stieltjes order suggests a natural angular-count question, but order- d positivity alone does not prove it. There is a rigorous small-radius starting point.

Proposition 12.1 (The small-radius angular count). *For a fixed positive index vector of depth d , and every sufficiently small $\rho > 0$, $\text{Im } F_s(\rho e^{i\theta})$ has exactly $d - 1$ simple zeros in $0 < \theta < \pi$.*

Proof. After division by $\rho^d \sin \theta$, its analytic coefficient expansion is

$$M_s U_{d-1}(c) + \sum_{n>d} c_n(s) \rho^{n-d} U_{n-1}(c), \quad c = \cos \theta.$$

The remainder tends to zero in $C^1([-1, 1])$ as $\rho \downarrow 0$. For instance $c_n \leq n^{d-1}$, $|U_{n-1}| \leq n$, and $|U'_{n-1}| \leq n^3$ on this interval. The latter bound follows by writing $U_{n-1}(\cos \theta)$ as a finite sum of cosines and using $|\sin(k\theta)| \leq k|\sin \theta|$. These polynomial bounds give uniform convergence with the first derivative. The polynomial U_{d-1} has precisely $d - 1$ simple roots $\cos(j\pi/d)$ in $(-1, 1)$ and no endpoint roots. Choose small disjoint neighborhoods of these roots where its derivative has a fixed nonzero sign. A sufficiently small C^1 perturbation has one simple root in each neighborhood and none on their compact complement. The case $d = 1$ has no roots. $\square$

Conjecture 12.2 (Full-radius strict-depth angular count). *For every positive index vector of depth d and every $0 < \rho \leq 1$, there are exactly $d - 1$ simple zeros of $\text{Im } F_s(\rho e^{i\theta})$ in $0 < \theta < \pi$.*

At depth one, the kernel $z/(1 - zu)$ has strictly positive imaginary part in the upper half-plane, so there are no angular zeros. The depth-two assertion is proved in this article. The arbitrary-depth assertion is a proposed research conjecture, supported here only by the small-radius proposition and the depth-one and depth-two cases; no exhaustive numerical higher-depth evidence is claimed.

12.4 Constructive measures and sharper asymptotics

The pushforward measure μ_s has a concrete positive construction, but an explicit one-variable density would improve quadrature and Padé computations. Conditional Dirichlet averages should

permit a piecewise-polynomial kernel in the x_j variables before integration against the logarithmic weight. It would be useful to derive endpoint asymptotics of this density with uniform constants as some s_j approach zero, and to sharpen (8.7) beyond the leading term. The boundary layer $\sum t_j \approx \log x$ interacts with the lower faces $t_j \approx 0$, so an all-orders expansion must track individual indices rather than only the total weight.

12.5 Colored periods, exact identities, and formalization

Other outer colors on the unit circle can destroy the positive sign of the double kernel. The right question is to characterize parameter and color sectors where a positive real kernel survives, rather than reuse a Gaussian sign argument formally. The normalization identities (9.2)–(9.3) may supply stable new coordinates for evaluating such colored functions. They do not resolve the existing S_6 period identity, whose finite mixed-reduction certificate remains a concrete arithmetic target.

The finite formalization core is small: the integer-root interval operations, binomial Euler weights, the rational root-kernel derivatives, and the Chebyshev sign checker. A subsequent analytic formalization can separate the compensated measure, the beta stochastic-order lemma, compact-slit continuation, and the large-negative-axis dominated convergence argument. The current scripts are reproducible exact arithmetic, not a Lean development.

Conclusion

The finite signed-measure boundary is real, but it is not the boundary of positive structure. Below it, compensation produces a finite positive first-difference measure and restores the analytic geometry that the old representation could not reach. This proves subcritical angular uniqueness, slit-plane nonvanishing, and normalized-radius increase. A different positivity mechanism—beta likelihood-ratio order at fixed total weight—settles the sharp critical Euler constant. Positive transport extends the evaluator to every positive pair, while ordered simplex kernels identify an exact Stieltjes order at all strict depths. Each statement has its own explicit scope, and the remaining period and supercritical phase questions retain their unresolved status.

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